Banavathu Damayanthi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.46
Jawahar Navodaya Vidhyalaya	Class XII (CBSE)	2022	92.20%
Jawahar Navodaya Vidhyalaya	Class X (CBSE)	2020	92.80%

EXPERIENCES

Qualcomm -- Software Engineer Intern

- Developed a **Python IMSDK pipeline** for real-time face detection using **YOLO models**, supporting both live camera streams and MP4 video inputs.
- Implemented **Face landmark detection post-processing**, extracting facial landmark coordinates from model-generated heatmaps.
- Developed **heatmap processing and coordinate transformations** to map facial landmarks from model-space to the original video/image resolution.
- Integrated and tested the pipeline within GStreamer-based multimedia workflows for both live-stream and file-based video processing.

PROJECTS

CPU Scheduling Simulator

- Developed a C++ simulator implementing **FCFS, SJF, SRTF, Round Robin, and Priority Scheduling** algorithms.
- Used **min-heaps and queues** for efficient process selection and scheduling, with heap operations.
- Generated **ASCII Gantt Charts** and computed **Waiting Time, Turnaround Time** to analyze scheduling performance.

Multithreaded Sudoku Validator

- Built a multithreaded C program to validate Sudoku puzzle solutions using POSIX threads (Pthreads).
- Created multiple threads to concurrently verify rows, columns, and subgrids, improving performance through parallelism.
- Implemented synchronization mechanisms to handle shared data and avoid race conditions .

Voice Activity Detection -- Voiced/Unvoiced Classification

- Developed a frame-based **Voice Activity Detection** system to classify speech segments as voiced or unvoiced.
- Used **autocorrelation analysis** to identify periodicity in speech frames and distinguish voiced speech from unvoiced speech.
- Applied **speech framing and windowing** and analyzed autocorrelation characteristics for frame-level classification and visualization.

Primal vs Dual Ridge Regression | Python, Numpy

- Implemented **Primal and Dual Ridge Regression** using NumPy and verified their **mathematical equivalence** through prediction comparison.
- Compared **runtime and computational efficiency** by varying the number of samples and features.
- Analyzed the dual formulation using a **Gram matrix**, highlighting its connection to **kernel methods.**

Face Data Analysis using PCA | Python, NumPy

- Applied **Principal Component Analysis (PCA)** to face data for dimensionality reduction and representation in a lower-dimensional feature space.
- Computed principal components and projected high-dimensional face data onto the selected components.
- **Reconstructed face data** from the reduced representation and analyzed the effect of the number of components on reconstruction quality.

SKILLS

- **Programming Languages:** Proficient in C ,C++ and Python.
- **Core Competencies:** Data Structures and Algorithms, Operating Systems, Foundations of Machine Learning, Object-Oriented Programming (OOPS).
- **Databases:** Relational Database Management System, MySQL
- **Software &Tools:** SolidEdge, Jupyter Notebook, NumPy, Pandas, Matplotlib.
- **Soft Skills:** Communication skills, Team Work, Time Management, Adaptability, Critical Thinking, Problem Solving.

RELEVANT COURSES

- **Programming & Computing:** Introduction to Programming, Scientific Computing, Computer-Aided Numerical Methods, Operating System -I, Operating System -II.
- **Mathematics & Analysis:** Probability, Statistics, Elementary Linear Algebra, Differential Equations,Transform Techniques, Calculus, Optimization Techniques.

SCHOLASTIC ACHIEVEMENTS

- **Excellence in SSC Examination:** Was the school topper in class 12 for securing second highest percentage of 92.2.
- **Recognized as Subject Topper:** Scored 100 out of 100 in Mathematics in CBSE 10 Board Examination.

EXTRACURRICULAR

- **Inter IIT Sports Meet (2023-2024):** Won Bronze Medal for Table Tennis .
- **CBSE merit certificate:** Recognized for outstanding performance in SSC(X) Mathematics .
- **All-Rounder Gold Medal:** Awarded for exceptional academic and extracurricular excellence in Grade XII at JNV .
- **SGFI Tournaments:** Represented school twice in Table Tennis at the national level.